

This prevents unauthorised photography/videography by using projectors and cameras that make any such photo/video blurred. This enhances the security of areas which don't allow photography. Complex embedded systems are used to develop high-end reliable devices for the use of military and defense department. US Marine Corps use a robotic system called Gladiator that can perform functions ranging from surveillance to assault and minimises the risk of marines in threat situations. Today governments are shifting from drones to small sized bots that can easily navigate. Recently, UK Ministry announced Black Hornet , world's first "nano-sized" surveillance system which is a toy sized helicopter that can fly unmanned over an area for about 30 minutes on battery charge. US army is also working on the idea of combining a number of robot snakes to form a tentacle system that has the ability to operate like a human hand and can be used for life-threatening tasks such as Bomb Disposal.

Careers in embedded systems

After 10-20 years, Embedded systems are going to be everywhere - On every floor, every wall, your coffee mug, public transports, cars, homes, offices, aeroplanes. Looking at the numerous possible applications it has and the awesome projects that are already in pipeline, it is safe to say that Embedded Systems will be one of the most sought after fields of study and hence would require a lot of talented minds.

And it is also important to note that as an Embedded systems Engineer, one would be dealing with some of the most advanced and complicated technologies while working on systems that would revolutionise the way we look and feel this world. So, choosing this field as a career option now would be a smart option to people who always were interested in Electronics and Electrical Engineering.

Study and pre-requisites

Embedded systems is more like an amalgamation of almost all cool tech of today - Software Programming, Digital Electronics, Mobile Computing, Wearable Computing, Augmented Reality and hence would require a base knowledge of all the above mentioned fields, especially Digital Electronics.

Top Universities around the globe including MIT, Carnegie Mellon, Stanford, Cornell, University of Tokyo, University of Cambridge provide both Undergraduate and Postgraduate courses on Embedded systems. In India, major technological institutions like IITs, NITs, IIITs provide Post-Graduate courses on Embedded systems with a pre-requisite knowledge/